

CSE321 Quiz 03 Set C

Time 20 minutes

[Q1 8 +Q2 2]

Q1. You are given a storage system with a total of 13000 blocks. Block size is 128 byte (2^7). the first block is Superblock. The next blocks are : 1 data bitmap block, 1 Inode bitmap block. The next ones are few INODES block and rest of them are data blocks

- Each INODE takes 32 (2^5) bytes of size. You have to accommodate at most 100 INODES in any given time. How many blocks will you need to allocate for the INODE block?
- After achieving that, if you wanted to directly go to the address space of INODE 52. What will be the block number and exact memory byte where it belongs. [logically. So no sector calculation needed]
- How many unique data blocks can you track for its allocation [free or not] given the properties mentioned.
- In INODE: 4 direct blocks, 1 single indirect and "X" double indirect blocks. Address space needed for each block is 4 bytes (2^2). You have to fit a file of max size 256 KB (2^{18}). What will be the value of X.

Q2.

- In the three criteria while writing a data: data bitmap update, inode update, data block write. In which point a crash has to appear for the situation to be labelled as "space leak"?
- In journaling, why is checkpointing necessary? [IN ONE LINE]